

Telescoping series - 5.2

Recall: $\int_a^b f'(x) dx = f(b) - f(a)$

Question: What is the analogue of the derivative?

Answer: differences! $\frac{a_{n+1} - a_n}{n+1 - n} = a_{n+1} - a_n$

If we look at partial sums of $(a_{n+1} - a_n)$

$$S_1 = a_2 - a_1, \quad S_2 = a_2 - a_1 + a_3 - a_2 = a_3 - a_1$$

$$S_3 = a_2 - a_1 + a_3 - a_2 + a_4 - a_3 = a_4 - a_1 \quad \dots$$

$$S_k = a_{k+1} - a_1 \quad \sum_{n=1}^{\infty} a_{n+1} - a_n = \lim_{k \rightarrow \infty} S_k = \lim_{k \rightarrow \infty} a_{k+1} - a_1$$

Sequences/series of differences are usually called 'telescoping series'.

Example: what is $\sum_{n=2}^{\infty} \frac{1}{n^2-n}$?

$$\rightarrow \frac{1}{n^2-n} = \frac{1}{n(n-1)} = \frac{A}{n} + \frac{B}{n-1} \Leftrightarrow 1 = A(n-1) + Bn$$

$$A = -1$$

$$B = 1$$

$$\sum_{n=2}^{\infty} \frac{1}{n^2-n} = \sum_{n=2}^{\infty} \frac{1}{n-1} - \frac{1}{n}$$

Differences of $a_n = -\frac{1}{n}$

$$S_k = \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{k-1} - \frac{1}{k} = 1 - \frac{1}{k}$$

$$\sum_{n=2}^{\infty} \frac{1}{n^2-n} = \lim_{k \rightarrow \infty} 1 - \frac{1}{k} = 1$$

Example: Evaluate $\sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n}\right)$.

$$= \sum_{n=1}^{\infty} (\ln(n+1) - \ln(n))$$

sequence of differences
for $a_n = \ln(n)$

$$\begin{aligned} S_k &= \cancel{\ln(2)} - \ln(1) + \cancel{\ln(3)} - \cancel{\ln(2)} + \dots + \ln(k+1) - \cancel{\ln(k)} \\ &= \ln(k+1) - \ln(1) = \ln(k+1) \end{aligned}$$

$$\sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n}\right) = \lim_{k \rightarrow \infty} \ln(k+1) = \infty$$

Series diverges.